LESSON 7.1

Explore Integers and Their Opposites

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use integers to represent quantities in everyday life, explain what 0 means in each situation, and name the opposite of an integer on the number line.

WHY THIS MATTERS The lowest elevation in North America is Badwater Basin in Death Valley: 86 meters below sea level. When we measure elevation, sea level is defined as 0. So 86 meters below sea level cannot be written as plain 86 — that would put it ABOVE the water. Numbers less than zero are negative numbers, written with a − sign, so the elevation of Badwater Basin is −86 meters. On a number line, −86 is 86 units to the LEFT of 0, while 86 is 86 units to the RIGHT of 0. Those two numbers are opposites.

REMEMBER

Explore Integers and Their Opposites

Integers & Opposites Counting numbers, their opposites, and 0 are integers. A negative integer is less than zero and is written with a − sign. Opposites are the same distance from 0, but on opposite sides of the number line.

1. Integers: Positive whole numbers, negative whole numbers, and zero (..., -2, -1, 0, 1, 2, ...).
2. Opposites: Numbers that are the same distance from 0 in opposite directions (e.g. -5 and 5).
3. The opposite of the opposite of a number is the number itself: -(-a) = a

Words: **Integer *(Número entero)*** **Opposite *(Opuesto)*** **Negative number *(Número negativo)*** **Positive number *(Número positivo)*** **Number line *(Recta numérica)***

1 **Circle the letter of the best answer. Show how you know.**

Simplify 8x + 5 − 3x.

(A) 5x + 5
(B) 5x
(C) 8x + 2
(D) 10x

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which pair are like terms?

(A) 4m and 9m

(B) 4m and 9

(C) 4m and $9m^2$

(D) 4m and 9n

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Simplify 2a + 7b + 5a.

(A) 7a + 7b

(B) 7a + 5b

(C) 12ab

(D) 14ab

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Simplify 6y − y.

(A) 5y

(B) 5

(C) 6

(D) 6y

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide what 0 meant in each situation, and how did that decide the sign of the integer?

SENTENCE STARTER In this situation 0 means ___. The quantity is ___ zero, so I represent it with a ___ integer, which is ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

To make a first down, a football team must gain at least 10 yards from where they started, a spot called the line of scrimmage. The line of scrimmage is 0: a gain moves the team above 0 and a loss moves the team below 0. During one series, a team had a 10-yard loss, a 4-yard gain, a 3-yard loss, and a 21-yard gain. What does 0 represent in this football situation?

(A) A play in which nobody touched the ball

(B) The line of scrimmage, the spot where the series started

(C) The end zone at the edge of the field

(D) The number of yards still needed for a first down

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

To make a first down, a football team must gain at least 10 yards from where they started, a spot called the line of scrimmage. The line of scrimmage is 0: a gain moves the team above 0 and a loss moves the team below 0. During one series, a team had a 10-yard loss, a 4-yard gain, a 3-yard loss, and a 21-yard gain. Which integer represents a 3-yard loss when the line of scrimmage is 0?

(A) -30
(B) -3
(C) 0
(D) 3

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

To make a first down, a football team must gain at least 10 yards from where they started, a spot called the line of scrimmage. The line of scrimmage is 0: a gain moves the team above 0 and a loss moves the team below 0. During one series, a team had a 10-yard loss, a 4-yard gain, a 3-yard loss, and a 21-yard gain. The four plays are -10, 4, -3, and 21. What is the team's net result?

(A) 12 yards gained
(B) 12 yards lost
(C) 18 yards gained
(D) 38 yards gained

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which situation is best represented by the integer -15?

(A) A diver 15 meters below the surface of the water
(B) A hill 15 meters above sea level
(C) A deposit of $15 into a bank account
(D) A football play with a 15-yard gain

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide what 0 meant in each situation, and how did that decide the sign of the integer?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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